◈ 1. Introduction

This project aims to analyze student data to develop an **"Expectation Decider" model**, which predicts whether a student will pass a competitive mathematics exam. The analysis is based on probability concepts, statistical techniques, and data visualization methods to identify the key factors that influence student performance.

◈ 2. Dataset Overview

The dataset consists of 200 student records with the following variables:

- **Study Hours**: Number of hours studied per week
- **Attendance**: Percentage of lecture attendance
- **Group Discussion**: Participation in group discussions (Yes/No)
- **Previous Test Score**: Marks obtained out of 100
- **Final Exam Result**: Pass/Fail outcome

◈ 3. Understanding Probability

➤ What is Probability?

Probability is a measure of the likelihood of an event occurring. Its value ranges from 0 to 1, where:

- 0 indicates an impossible event
- 1 indicates a certain event

➤ Key Terminology

- **Experiment**: Selecting a student
- **Outcome**: Pass or Fail
- **Event**: A student passes the exam
- **Sample Space**: All possible students in the dataset

➤ Examples from Dataset

- Probability that a student passes the exam

- Probability that study hours exceed 10 hours per week
- Probability that attendance is greater than 80%

◈ 4. Types of Probability

➤ Empirical Probability

Empirical probability is calculated using observed data:

$$P(\text{Pass}) = \frac{\text{Number of students who passed}}{\text{Total number of students}}$$

➤ Theoretical Probability

Theoretical probability assumes equal likelihood of outcomes:

$$P(\text{Pass}) = \frac{1}{2}$$

◈ 5. Random Variable and Probability Distribution

Let **X** be the number of students who pass out of 3 randomly selected students.

$$X = 0,1,2,3$$

Since each student either passes or fails, this follows a **binomial distribution**.

➤ Binomial Distribution Formula

$$P(X = k) = \binom{3}{k}(0.55)^k(0.45)^{3-k}$$

Where:

- $p = 0.55$(probability of passing)
- $1 - p = 0.45$
- $n = 3$

➤ **Probability Distribution Table**

X (Students Passing)	Calculation	Probability
0	$(0.45)^3$	0.0911
1	$3 \times (0.55)(0.45)^2$	0.3341
2	$3 \times (0.55)^2(0.45)$	0.4084
3	$(0.55)^3$	0.1664

➤ **Mean (Expected Value)**

$$\text{Mean} = np = 3 \times 0.55 = 1.65$$

➤ **Variance**

$$\text{Variance} = np(1-p) = 3 \times 0.55 \times 0.45 = 0.7425$$

◈ **6. Understanding Relationships**

➤ **Given Values**

- $P(\text{Pass}) = 0.55$
- $P(\text{GD AND Pass}) = 0.34$
- $P(\text{Pass} \mid \text{GD}) = 0.66$

➤ **Interpretation**

The conditional probability of passing given participation in group discussions (0.66) is higher than the overall probability of passing (0.55). This indicates that students who participate in group discussions have a higher likelihood of passing the exam.

Thus, group discussion participation has a **positive impact on student performance**.

➤ **Independence Check**

For two events to be independent:

$$P(A \cap B) = P(A) \times P(B)$$

Let:

- $A =$ Pass
- $B =$ Group Discussion

Step 1: Calculate $\boldsymbol{P(GD)}$

$$P(GD) = \frac{0.34}{0.66} \approx 0.515$$

Step 2: Check Condition

$$P(Pass) \times P(GD) = 0.55 \times 0.515 = 0.283$$

Since:

$$0.34 \neq 0.283$$

the events are **not independent**.

➤ **Final Conclusion (Relationship)**

- The events are **dependent**
- The events are **not mutually exclusive** since they can occur together

Therefore, participation in group discussions significantly influences the probability of passing the exam.

◈ **7. Bayes Theorem**

➤ **Given Data**

- $P(HighAttendance \mid Pass) = 0.7$

- $P(HighAttendance \mid Fail) = 0.4$
- $P(HighAttendance) = 0.6$

➤ **Step 1: Find Prior Probabilities**

$$P(Pass) = 0.55, P(Fail) = 0.45$$

➤ **Step 2: Apply Bayes Theorem**

$$P(Pass \mid HighAttendance) = \frac{P(HighAttendance \mid Pass) \cdot P(Pass)}{P(HighAttendance)}$$

➤ **Step 3: Substitute Values**

$$P(Pass \mid HighAttendance) = \frac{0.7 \times 0.55}{0.6}$$

➤ **Step 4: Final Result**

$$P(Pass \mid HighAttendance) = 0.642$$

➤ **Interpretation**

The probability that a student passes given high attendance is approximately **64.2%**, which indicates that attendance plays a significant role in improving student performance.